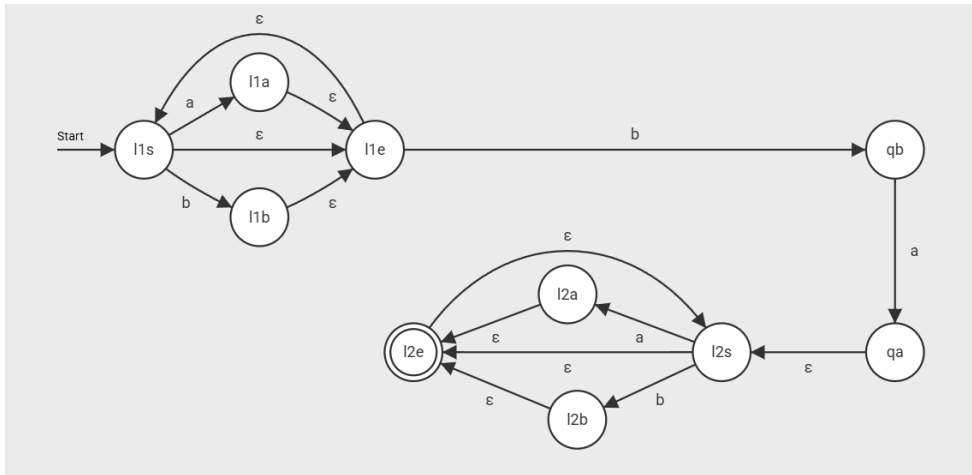


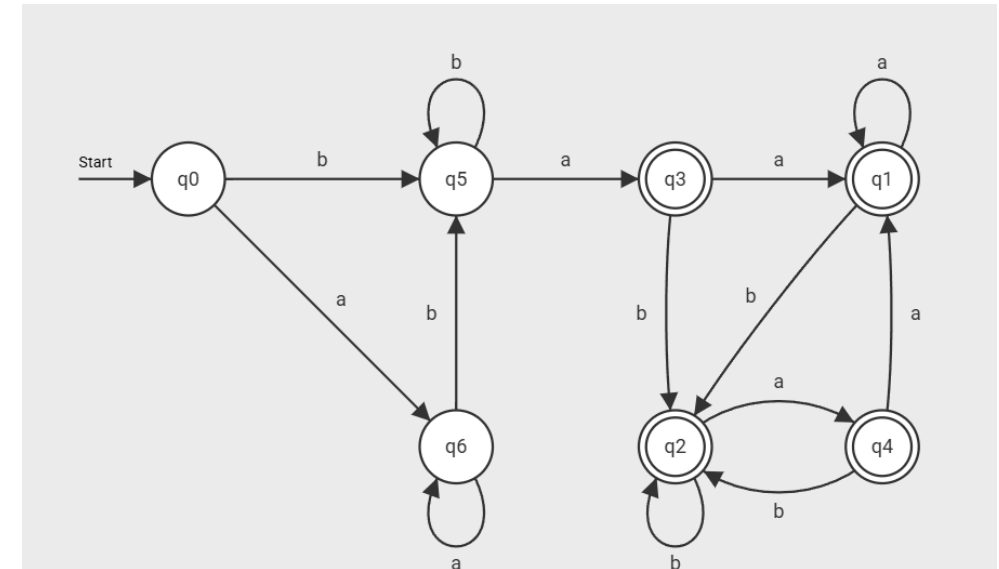
CC02

Exercise 2.1

2.1.1: NFA from RegEx



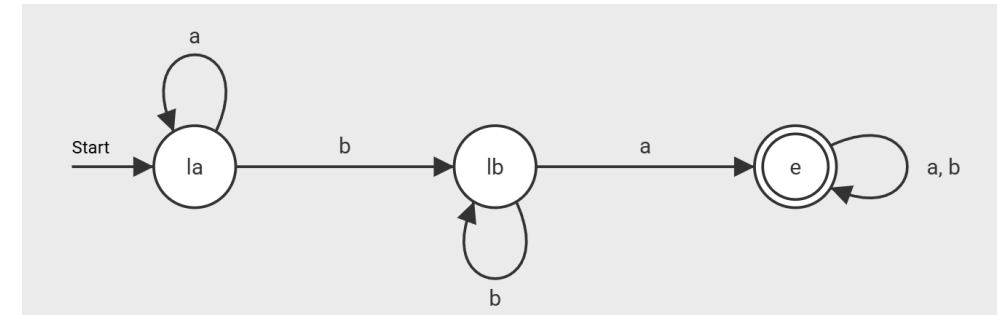
2.1.2: DFA from NFA



$q3$ can only be reached, if ba is part of the word. Until $q3$ $(a|b)^*ba$ is checked while the second b - a -loop is only checked after $q3$.

2.1.3: minimized DFA

Of course such a DFA can be minimized.



lb performs the same function as $q5$ does in 2.1.2; then e performs the same function as $q1, q2, q3$ and $q4$.